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Case report

Systematic forensic identification of a homicide by brodifacoum poisoning: A case report

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ABSTRACT

Brodifacoum, a long-acting anticoagulant rodenticide, exerts inhibitory effects on blood coagulation factor synthesis, leading to abnormal clotting function and potential fatality. Poisoning of accidental exposure to brodifacoum is not rare, but death from brodifacoum poisoning can be largely avoided with timely and long-term effective treatments, consequently, forensic data on fatalities due to brodifacoum poisoning may be limited. This paper presents a case of brodifacoum-induced homicide, detailing the medial records and pathological changes observed in multiple organs. Furthermore, t liquid chromatography–tandem mass spectrometry (LC-MS/MS) was employed to measure the concentrations of brodifacoum in blood, hydropericardium, and urine 40 days post-poisoning (1 day post-mortem), yielding values of 0.097 μg/mL, 0.089 μg/mL, and 0.007 μg/mL respectively. The aim of this article is to contribute towards forensic identification of fatalities resulting from brodifacoum poisoning while also serving as a reference for clinical diagnosis in similar cases.

1. Background

Brodifacoum irreversibly binds to the vitamin K epoxide reductase complex, inhibiting the conversion of vitamin K from epoxide to hydroquinone. This hinders the synthesis of vitamin K-dependent coagulation factors (VII, IX and X), protein C and protein S, ultimately leading to coagulation dysfunction.[1] Accidental exposure to brodifacoum can result in poisoning, which has been reported in some cases.[2–4] However, effective medical treatments for brodifacoum poisoning are available and even if there is few detection method in most hospitals, experienced doctors can accurately determine appropriate treatment based on symptoms and auxiliary examination results,[5–7] reducing fatalities caused by brodifacoum poisoning. Consequently, forensic data related to deaths caused by brodifacoum poisoning is scarce including pathological manifestations of multiple organs and detection methods for postmortem samples containing brodifacoum.

Therefore, this paper presents a case of homicide in which a victim was poisoned with brodifacoum but received medical treatment for only 5 days before succumbing on day 39 after ingestion. Detailed descriptions are provided regarding medical records and pathological changes observed in multiple organs. Additionally, concentrations of brodifacoum were measured using LC-MS/MS in blood samples as well as hydropericardium and urine collected on the first day after death with an aim to provide forensic diagnostic evidence for similar cases.

2. Case presentation

In order to stop suffering domestic violence, a wife who had been abused for more than 20 years and was refused a divorce added a certain amount of brodifacoum to the pot while cooking rice to kill her 65-year-old husband. Approximately 24 days after brodifacoum ingestion, the man noticed blood in his urine prompting him seek medical assistance at a hospital two days later when hematuria persisted. Over the next 4 days, he underwent various clinical examinations and treatments until

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his discharge on day 30 post-poisoning, when the hematuria had ceased. Unfortunately, on the morning of 9 days after discharge, the man vomited blood and subsequently fell into a coma before passing away at noon on the same day. The total duration of poisoning was 39 days. During his 5-day hospitalization, coagulation tests were primarily conducted (Table 1). Concurrently, intravenous administration of vitamin K1 served as the main treatment.

3. Autopsy findings

3.1. Macroscopic examination

An autopsy was performed one day after death. The victim exhibited pale skin with dark purple lividity and multiple subcutaneous bleedings were observed: left hand back (6.0 cm × 4.5 cm), right hand back (3.0 cm × 1.8 cm), right lower limb (with muscle hemorrhages) and right instep (42.0 cm × 18.0 cm).

The brain weighed 1390 g and revealed a subdural hematoma measuring 8.0 cm × 5.0 cm on the top-right side along with extensive subarachnoid hemorrhage (SAH) in both cerebral hemispheres, more severe on the left side; small subarachnoid hemorrhages were also found in cerebellar vermis and right posterior cerebellar lobe (Fig. 1A). A sagittal incision of the brain exposed a hematoma in the left frontal parenchyma measuring 3.5 cm × 3.0 cm × 2 .5cm (Fig. 1B).

Both lungs weighed 1430g, exhibiting scattered hemorrhagic spots on their surface along with edema, emphysema, and bleeding in their lower lobes' parenchyma. The heart weighed 330g, and the left anterior descending branch showed gradeIIatherosclerotic plaque formation while no other abnormalities were detected.

The liver weighed 1120g, measuring 22.0cm × 7.0cm × 6.0cm without any obvious gross abnormalities. The spleen weighed 430g, measuring 16.0cm × 11.0cm × 6.0cm and displayed swelling. Bilateral kidneys weighing 320g had soft tissue hemorrhage around the right kidney. Gastric mucosa presented scattered hemorrhagic spots. No abnormalities were found in other organs.

3.2. Microscopic examination

Extensive bleeding was observed in the skin and muscle tissue of the victim's right lower limb and instep. Microscopic examination revealed multiple subarachnoid hemorrhages in both cerebral hemispheres, in particular, the left frontal pole was more serious, in which neutrophil-dominated inflammatory cell infiltration was seen (Fig. 2A), indicating recent occurrence before death. In addition, cerebral edema and ischemic hypoxic changes were also present in some neurons.

Table 1
Results of coagulation tests during the patient's hospitalization.

Indicators of coagulation tests	Days of the hospitalization (d)					
	1*	1#	1&	2*	2&	4*
APTT (20–40 S)	137.99 ↑	133.77 ↑	56.42 ↑	48.20 ↑	44.33↑	41.94 ↑
PT (10.7–14.0 S)	>100 ↑	>100 ↑	21.66 ↑	16.58 ↑	15.05 ↑	17.09 ↑
TT (13–21 S)	17.50	16.49	17.99	17.90	17.52	16.69
INR (0.8–1.2)	>7.99 ↑	>7.99 ↑	1.62↑	1.25 ↑	1.14	1.29 ↑
FIB (2–4 g/L)	2.94	2.83	2.54	2.51	2.40	2.91
D-Dimer (0–0.55 mg/L)	1.33 ↑	0.36	0.37	0.22	0.15	0.36

Note: The numbers in the parentheses represent the reference range for each parameter; "↑" represent the value is higher than the reference range; ">" represents the result has exceeded the quantitative upper limit; "*", "#" and "&" represent the time of blood was collected in the morning, noon and afternoon, respectively. (Abbreviations: APTT: activated partial thromboplastin time; PT: prothrombin time; TT: thrombin time; INR: international normalized ratio; FIB: fibrinogen.)

Diffuse massive hemorrhages, edema and emphysema were found microscopically in both lungs (Fig. 2B). Scattered fibrosis was observed in the interstitial myocardium of the left ventricle. The left anterior descending (LAD) branch exhibited secondary atherosclerosis with minimal mononuclear lymphocyte infiltration and plaque bleeding (Fig. 2C).

Under microscopic examination, liver cells displayed edema and multiple foci of patchy necrosis. Some hepatic sinusoids were disrupted with bleeding or small hematomas, while others contained mononuclear lymphocytes, and fibrosis (Fig. 2D). The central artery intima of the spleen showed thickening with hyaline degeneration, accompanied by obvious splenic sinus congestion leading to small hematoma formation (Fig. 2E). Some glomeruli exhibited hyaline changes along with interstitial fibrosis characterized by mononuclear cell aggregation; most renal tubule epithelial nuclei disappeared and protein tubules had been formed within certain renal tubules (Fig. 2F). Scattered spotty bleeding occurred in gastric mucosa, accompanied by a small amount of inflammatory cell infiltration. No significant positive pathological changes were found in other organs or tissues.

4. Toxicological analyses

4.1. Sample preprocessing

Blood, hydropericardium and urine samples were all treated by protein precipitation as follows: 100 μL of sample was collected and added with 10 μL of internal standard solution (warfarin-d5, 1 μg/mL). Then, 400 μL of acetonitrile (MS grade, SCRC, China) was added into the mixture and vortexed for 1 minute. After centrifugation at 700×*g* (DLAB, China) for 3 minutes, the supernatant was collected, filtered by 0.22μm filter membrane (Beyotime, China) and then subjected to testing on the LC-MS/MS system.

4.2. LC conditions

LC-MS/MS8045 system (SHIMADZU, Japan) was used for testing. Samples were separated on an Agilent Eclipse Plus C18 column (100 mm × 2.1 mm × 3.5 μm, Agilent, USA). The mobile phases consisted of: (A): 10mmol/L ammonium acetate (ALADDIN, China) plus 5 % methanol (SCHARLAU, Spain); (B):100 % methanol (SCHARLAU, Spain) (All the above reagents were of MS grade), and delivered through the column (temperature +25 °C) at a flow rate of 200 μL/min. The elution procedure lasted for 14 min, which started at 20 % B for 1.5 min, raised to 80 % B in 1.5 min and held for 2 min. Phase B then rose to 95 % in 0.5 min and held for 3.5 min, returned to 20 % in 1 min, then 20%B held for 4 min, to equilibrate the column before the next injection.

4.3. MS conditions

Electrospray ionization (ESI) source in negative ionization mode was adopted to collect MS data, with multiple reaction monitoring (MRM) scan mode. The qualitative ions of the brodifacoum were 521.10 > 135.10 and 521.10 > 187.15, which the 521.10 > 135.10 played a quantitative role (Fig. 3).

4.4. LC-MS/MS results

Brodifacoum was successfully detected in the sample of blood, hydropericardium and urine, with quantitative results of 0.097 μg/mL, 0.089 μg/mL and 0.007 μg/mL. In addition, ethanol and other common toxic drugs were not detected in the three samples. The chromatograms of brodifacoum and warfarin-d5 (internal standard) of the blood sample were presented as an example (Fig. 3).

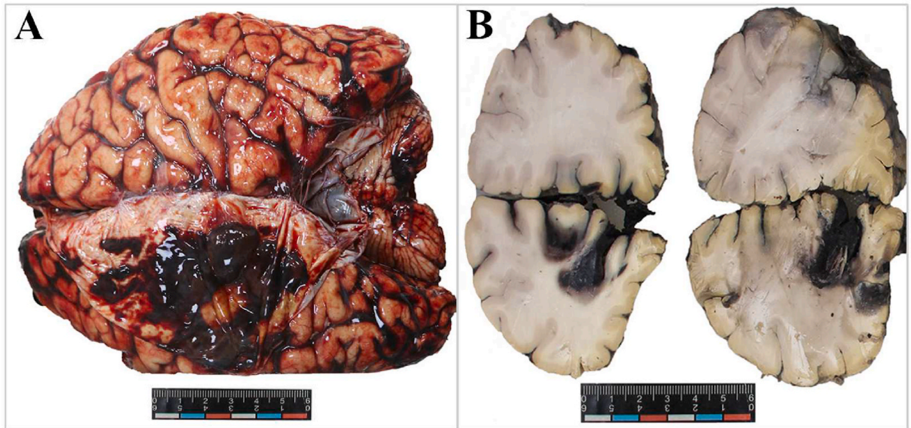

Fig. 1. The hemorrhages in different organs or tissues. (A: Subarachnoid hemorrhage in both cerebral hemispheres and cerebellum; B: Hematoma in the left frontal parenchyma).

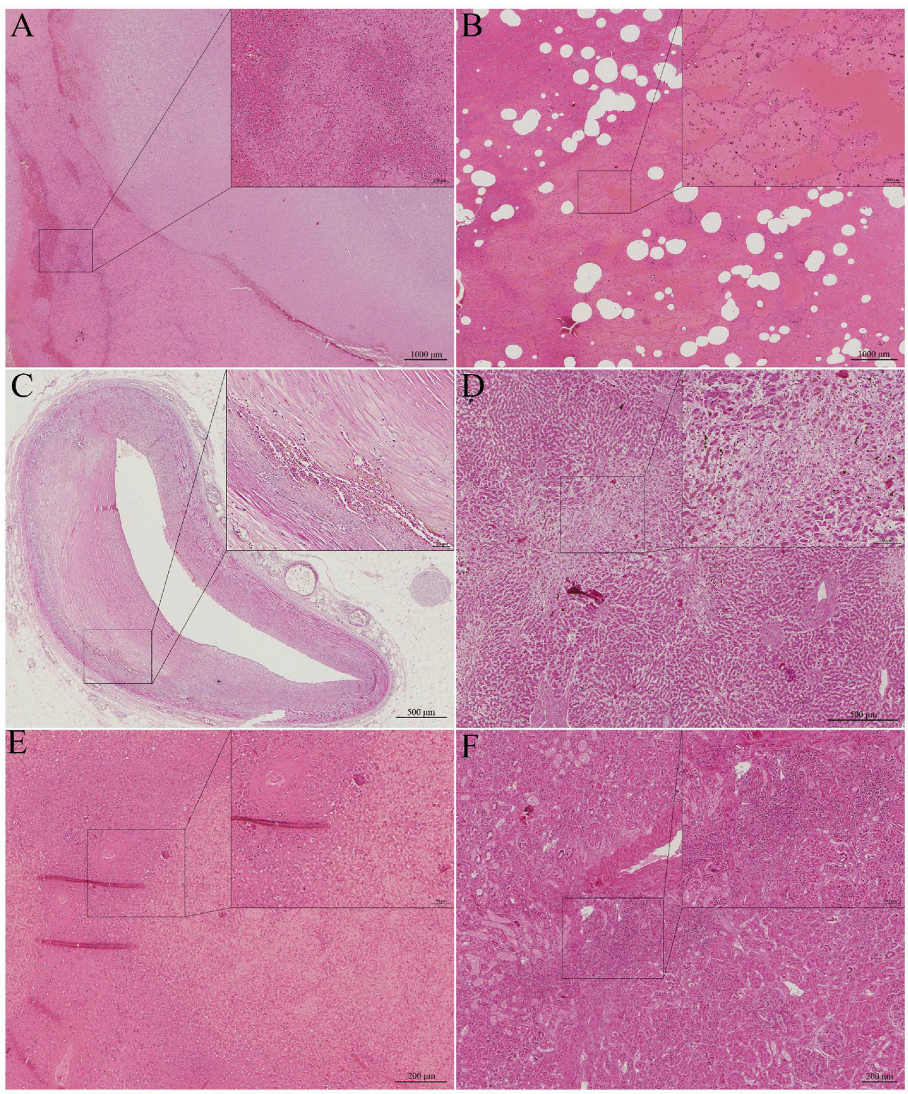

Fig. 2. Pathological changes of multiple organs microscopically.. (A: SAH in the left frontal pole with inflammatory cell infiltration; B: Hemorrhages, edema and emphysema in pulmonary alveoli. C: Secondary atherosclerosis in the LAD with plaque bleeding; D: Hepatocyte necrosis, hemorrhage and fibrosis; E: Hyalinosis of central splenic artery and splenic hematoma; F: Renal interstitial fibrosis, inflammatory cell infiltration and protein tube formation.)

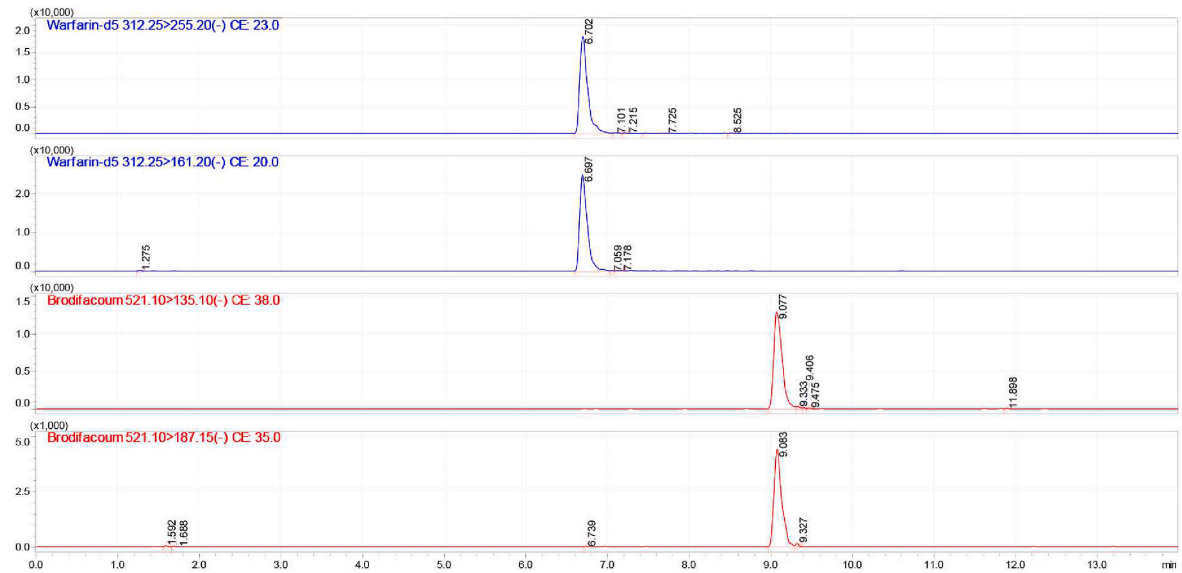

Fig. 3. The chromatogram of warfarin-d5 and brodifacoum of the blood sample.

5. Cause of death

Based on the medical records, forensic pathological examination, and toxicological analysis, a comprehensive analysis indicates that the deceased individual met the criteria for death due to toxicity from brodifacoum, resulting in widespread bleeding of multiple organs (especially multiple hemorrhages in the brain) and ultimately led to fatal acute failure of the central nervous system.

6. Discussion

Brodifacoum can obstruct the synthesis of vitamin K-dependent coagulation factors, leading to coagulation dysfunction. There is typically a latency period of 3–9 days after intake due to pre-existing clotting factors.[8] However, in this case, hematuria was observed 22 days after intake, which exceeds the reported latency. We believe that lower doses of brodifacoum may incompletely inhibit clotting factor synthesis, resulting in a longer incubation period.[9,10] After this incubation period, patients usually exhibit various bleeding manifestations such as hematuria, vaginal and rectal bleeding, gastric and pulmonary hemorrhage,[10–12] in which hematuria is often the earliest and most significant clinical symptom of coagulation dysfunction,[13] consistent with what was observed in this case. Additionally, high results of APTT (activated partial thromboplastin time), PT (prothrombin time), and INR (international normalized ratio) on admission further suggest coagulation dysfunction.

However, after treatment of vitamin K_1, these elevated outliers gradually decreased and approached the normal range by the fourth day of admission, when no hematuria was observed. Despite this improvement in clotting factors, the patient still succumbed to death. We speculate that several reasons may have contributed to this outcome:

Firstly, there was an initial failure to correctly identify the cause of coagulation dysfunction. In this case, due to a lack of diagnostic methods for common rodenticides in the hospital, the etiology of the patient's condition was not accurately diagnosed initially. To facilitate clinical diagnosis and forensic identification in similar cases, we proposed a method utilizing LC-MS/MS for detecting brodifacoum in human samples after pre-treatment with protein precipitation. This method successfully quantified brodifacoum concentrations in the deceased's blood (0.097 μg/mL), pericardial effusion (0.089 μg/mL), and urine (0.007 μg/mL). Although the concentration of brodifacoum in blood in this case was lower than those reported in some previous literatures, such as 0.681 μg/mL on day 43[8], 0.170 μg/mL on day 30[14], it was still higher than the other reported result (0.032 μg/mL on day 9[15]), and was within the ranged from 0.030 to 0.786 μg/mL[16] of fatal poisonings brodifacoum levels in blood samples. Thus, we concluded that this patient had a lethal concentration of brodifacoum in their blood. However, since the container containing the brodifacoum was not found successfully, and the weight of the rice containing brodifacoum which the victim ate was also uncertain, it was a pity that the intakes cannot be inferred, as the reference 8[8].

Secondly, the unclear diagnosis led to inappropriate follow-up treatments. Brodifacoum primarily inhibits the synthesis of vitamin K-dependent coagulation factors and has a long half-life, ranging from 20 to 130 days.[17] Even though vitamin K_1 supplementation can temporarily restore some coagulation function indicators to normal, once the treatment is stopped, the residual brodifacoum will continue causing coagulation dysfunction. Therefore, we believed that a long-term and continuing treatment was essential to avoid death, which seemed reasonable because of the existence of cases in which victims were alive after prolonged treatments even though poisoned (e.g., treatments with phytonadione last 114 days,[18] treatments with vitamin K last 4 months,[19] and the average treatment courses of 168 days in 174 poisoning cases[1]). Hence, in this case, despite close-to-normal coagulation function indicators after four days of treatment, discontinuation ultimately resulted in death. Regarding non-lethal concentrations of brodifacoum, it has been reported that complete symptom disappearance and a blood concentration below 0.010 μg/mL will not cause coagulation dysfunction,[10,14] which could provide reference for clinical practice.

As for the postmortem pathological manifestations of the patient, in addition to the frequently reported multi-organ hemorrhages mentioned above, this case also presented rarely reported cerebral subarachnoid and intracerebral hemorrhages, as well as intracerebral hemorrhage in the left anterior descending branch plaque. Furthermore, toxic liver and kidney diseases were diagnosed. Notably, there was evident subarachnoid hemorrhage in both cerebral hemispheres and part of cerebellum without inflammatory cell infiltration. Conversely, inflammatory cell infiltration was found in both left frontal parenchymal hemorrhage and left anterior descending plaque hemorrhage, indicating a shorter bleeding time before death. The amount of parenchymal bleeding and inflammatory cells was larger, indirectly suggesting central nervous system failure as the cause of death. Additionally, varying degrees of fibrosis accompanied by mononuclear lymphocyte infiltration were observed in the liver and kidney of the deceased individual along with liver necrosis, bleeding, and renal protein tubules; these findings further suggested chronic poisoning possibility.

In conclusion, we present a detailed description of a homicide involving brodifacoum poisoning including antemortem medical records and postmortem pathological changes that are rarely reported elsewhere. Moreover, we propose a method for quantifying brodifacoum concentration in liquid samples which can yield quantitative results for blood analysis as well as pericardial effusion and urine analysis from deceased individuals. This article aims to offer a clinical diagnostic approach and forensic identification method for similar cases while providing valuable insights into estimating lethal concentrations of brodifacoum poisoning.

Informed consent

Informed consent forms have been obtained from relatives of the deceased.

Declaration of competing interest

All authors have agreed and declared that there was no conflict of interest.

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